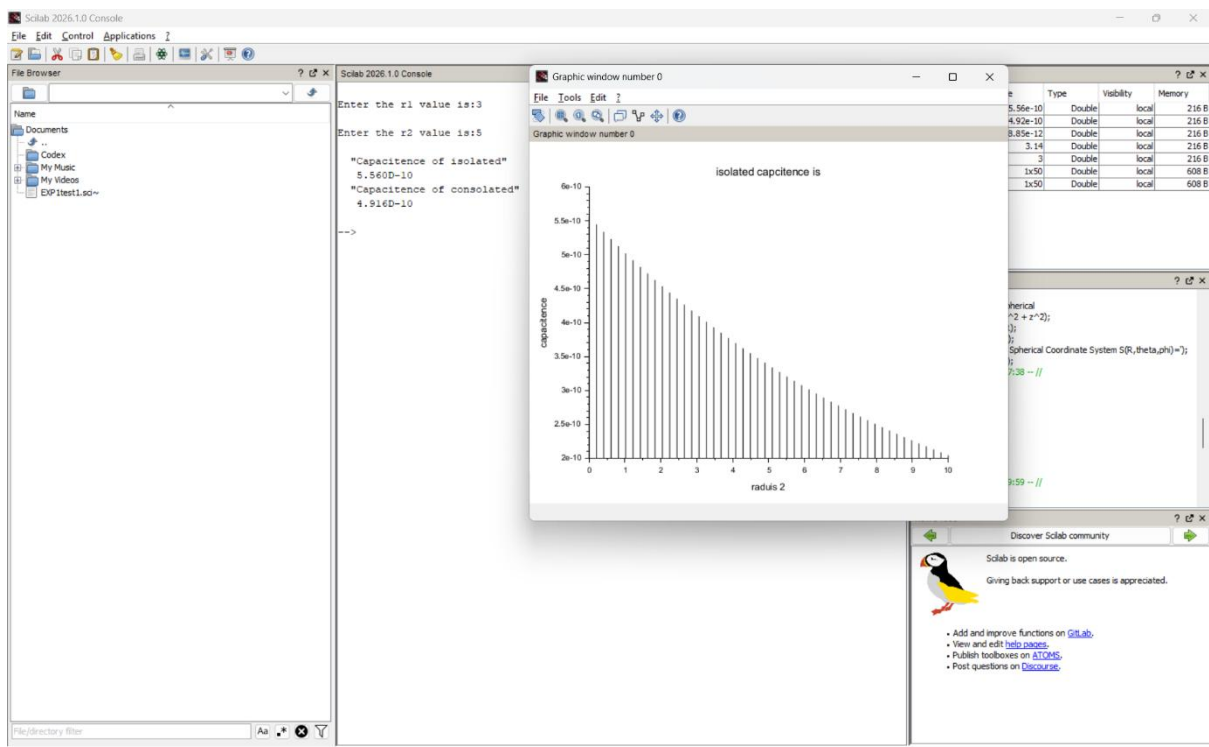


EXP 7: Determination and plotting of Capacitance of isolated and concentric sphere with different inner and outer radius and dielectric constant of the dielectric mediums.

```

1 clear;
2 clear;
3 pi=3.14;
4 e=8.85410^-12;
5 r1=input("Enter the r1 value is:");
6 r2=input("Enter the r2 value is:");
7 c1=(pi*e*r2);
8 disp("Capacitance of isolated",[c1]);
9 r2=input("Enter the r2 value is:");
10 c2=4*pi*e*(r2^2);
11 plot(2:9,r2,c2);
12 xlabel("radius-2");
13 ylabel("capacitance");
14 title("Isolated capacitance is");
15 co=(pi*e*(r1^2+r2^2))/(r2-r1);
16 disp("Capacitance of consolated",[co]);

```



Exp 7:-

$$r_1 = 3 \text{ cm}$$

$$r_2 = 5 \text{ cm}$$

(i) capacitance of isolated sphere

$$C_i = 4\pi\epsilon r_2$$

$$= 4 \times 3.14 \times 8.854 \times 10^{-12} \times 5$$

$$= 12.56 \times 8.854 \times 10^{-12} \times 5$$

$$= 5.560 \times 10^{-10}$$

$$C_i = 5.560 \times 10^{-10} \text{ F}$$

(ii) capacitance of concentric sphere

$$C_o = \frac{4\pi\epsilon (r_1 r_2)}{r_2 - r_1}$$

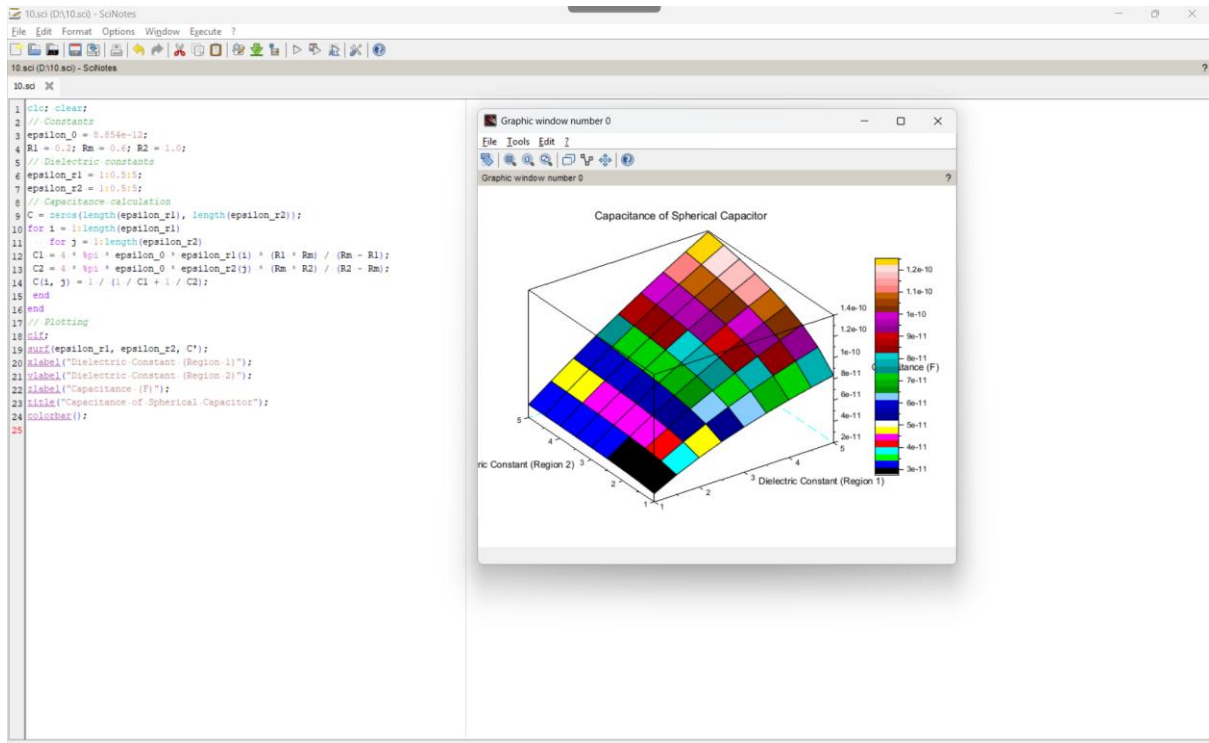
$$C_o = \frac{4 \times 3.14 \times (8.854 \times 10^{-12}) \times 15}{5 - 3}$$

$$= \frac{12.56 \times 8.854 \times 10^{-12} \times 15}{2}$$

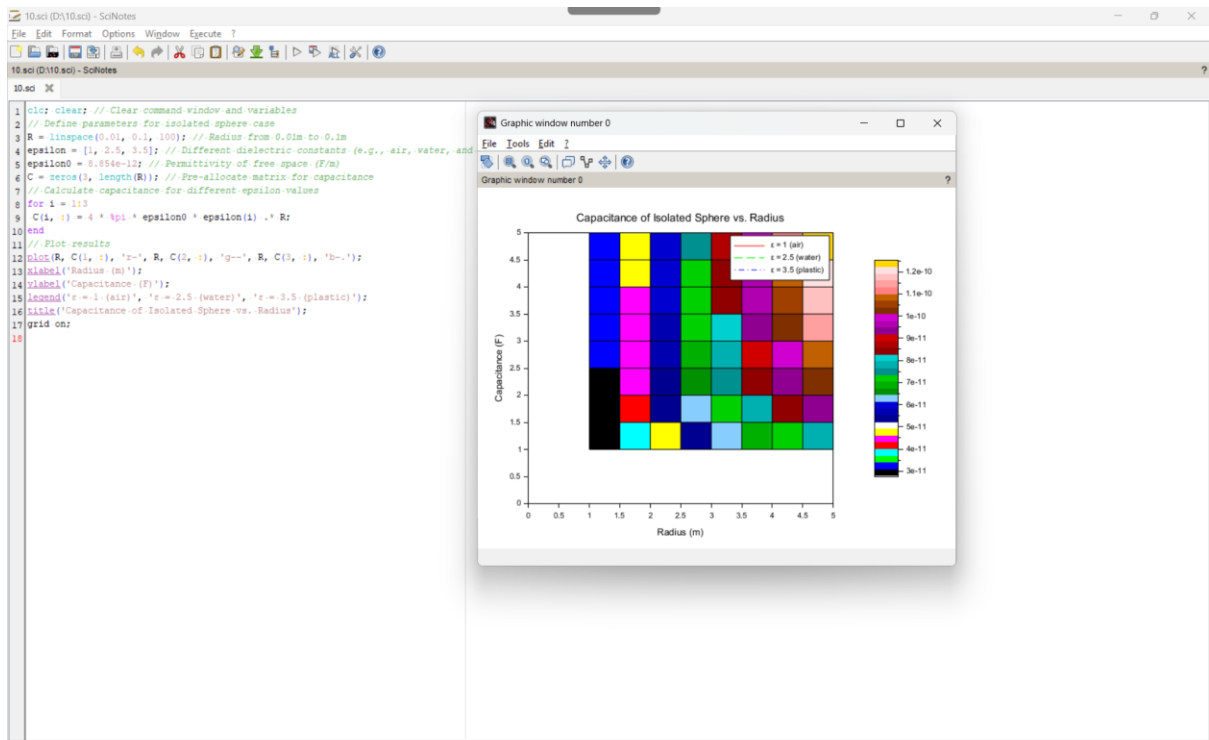
$$= 4.916 \times 10^{-10}$$

$$C_o = 4.916 \times 10^{-10} \text{ F}$$

CASE 1:-CONCENTRIC SPHERICAL CAPACITOR WITH DIFFERENT DIELECTRIC MEDIUMS



CASE 2: ISOLATED SPHERE



CASE 3: CONCENTRIC SPHERICAL CAPACITOR

